

VISHWAS KOTHARI

M.S. COMPUTER SCIENCE | ML EVALUATION | DATA SYSTEMS | RESPONSIBLE AI

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PROFILE

M.S. Computer Science student (May 2027) who builds and evaluates ML and data systems end to end — from research pipelines and optimization workflows to reproducible evaluation harnesses, fairness diagnostics, and user-facing applications, with an emphasis on verifying that results hold up before they ship.

EDUCATION

UNIVERSITY OF COLORADO BOULDER

Boulder, CO | Aug 2025 – May 2027

Professional Master's in Computer Science | GPA: 3.74/4.0

Relevant Coursework: Neural Networks & Deep Learning, Theory of Machine Learning, Design & Analysis of Algorithms, Database Systems, Linear Programming

VELLORE INSTITUTE OF TECHNOLOGY

Vellore, India | Aug 2021 – Jun 2025

B.Tech in Information Technology | GPA: 8.49/10.0

EXPERIENCE

SPACE APPLICATIONS CENTRE, INDIAN SPACE RESEARCH ORGANISATION (ISRO)

Ahmedabad, India

Research Intern, Biological Oceanography Division | Jan 2025 – Jul 2025

- Compared local and global optimization by implementing Levenberg–Marquardt and custom Particle Swarm Optimization on NASA NOMAD; LM reached $R^2 = 0.987$ on reflectance reconstruction and log-space $R^2 = 0.879$ on chlorophyll validation ($n = 105$).
- Automated OCM-3/MODIS validation with a Python pipeline (SciPy, NumPy, Pandas, xarray) for reprojection, resampling, date-based pairing, pixel-level colocation, area-of-interest analysis, and NetCDF/CSV outputs.
- Standardized research handoff through four agreement metrics (RMSE, MAE, R^2 , bias), six diagnostic visualization types, a 44-page technical report covering both projects, and presentations to cross-division research staff.

SELECTED PROJECTS

FINAGENT-EVAL | GitHub

Python, PyTorch, vision-language models, GitHub Actions, pytest, Streamlit | 2026

- Built a reproducible evaluation harness benchmarking five models on 397 financial-document questions across four categories, with schema-normalized grading and 1,000-sample bootstrap intervals.
- Investigated 151 labeled failures behind the leading aggregate score and found only 4 of 65 numerical-reasoning questions correct, exposing reliability limits that headline accuracy concealed.
- Shipped 12 CI-gated tests, a seven-tab evaluation dashboard, full documentation, and a reproducible analysis notebook for independent verification.

FINANCIAL DOCUMENT VQA | GitHub

Python, PyTorch, LayoutLMv3, Donut, Pix2Struct, LoRA | 2026

- Curated a 397-question SEC-filing benchmark from 79 images across 10 companies and four task types; compared four document AI architectures, observed a 63–83% drop from reported generic benchmark performance, and identified a ranking reversal on dense tables.
- Raised held-out ANLS from 0.097 to 0.134 with finance-specific LoRA trained on 307 examples and evaluated on 90 questions from three unseen companies; documented that exact match did not improve and generic fine-tuning caused negative transfer.

XAI CREDIT LENS | GitHub

Python, Streamlit, Scikit-learn, XGBoost, LightGBM, SHAP, LIME, DiCE, Optuna | 2026

- Built a credit decision-support prototype on 30,000 records, comparing three tree-based models with Optuna and stratified five-fold cross-validation; the selected XGBoost model achieved 0.782 test ROC AUC.
- Integrated SHAP, LIME, and DiCE to surface decision drivers and prototype counterfactual scenarios for individual predictions.
- Implemented fairness diagnostics across four protected attributes (stored disparate-impact ratios: 0.960–1.029) and built prototype governance-report modules mapped to ECOA and EU AI Act documentation requirements.

IMPLICIT BIAS OF ADAM VS. SGD | GitHub

Python, PyTorch, CVXPY, NumPy, controlled robustness testing | 2026

- Compared learned directions with exact convex max-margin solutions across five seeds, reaching cosine similarities of 0.990 for gradient descent and 0.988 for Adam.
- Stress-tested the two training recipes under a flipped 99%-to-1% spurious correlation, measuring a 30.2-point out-of-distribution accuracy drop for SGD versus 13.6 for Adam.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Java, C/C++, NumPy, Pandas, SciPy, xarray

ML & Analysis: PyTorch, Scikit-learn, XGBoost, LightGBM, CVXPY, SHAP, LIME, DiCE, LoRA, statistical analysis, Matplotlib, Seaborn, Plotly

Engineering & Applications: Git, GitHub Actions (CI), pytest, Linux, Jupyter, Streamlit, Gradio, $\LaTeX$

PUBLICATION, CERTIFICATION & SERVICE

Publication: Kothari, V. and Krishwanth, B. (2025). *Empowering Survivors: Ethical AI for Countering Violence Against Women*. Springer.

Certification: AWS Certified Cloud Practitioner — 970/1000 (2024).

Peer Reviewer: *Public Health and Social Sciences and Humanities Open* (Elsevier, 2026 – Present).